

Udacity Programming Assignment 2 // Angela Gross // Parallel Programming

The screenshot shows a web browser window with the Udacity Classroom interface. The browser's address bar displays the URL: `https://www.udacity.com/course/viewer#!/c-cs344/l-77202674/e-80151292/m-81489114`. The browser's tab bar includes several open tabs: 'Personal To Dos', 'Assignment', 'Classroom - Uda', 'How I got (a hair', 'MightyText', 'CUDA Program', '557*313 - Googl', and 'Trello'. The browser's bookmark bar shows various bookmarks including 'Apps', 'Xsede', 'XSEDE User Portal', 'Parallel Deadlines', and 'Parallel Programmin...'. The Udacity Classroom interface has a left sidebar with navigation links: 'DASHBOARD', 'CLASSROOM', 'MATERIALS', 'DISCUSSION', and 'OVERVIEW'. The main content area is titled 'Lesson 2 - GPU Hardware and Parallel Communication Patterns' and 'Problem Set #2'. It features a file explorer with four files: 'student_func.cu' (selected), 'reference_calc.cpp', 'main_reference.cpp', and 'HW2_reference.cpp'. Below the file explorer is a code editor showing the contents of 'student_func.cu'. The code is a CUDA kernel for blurring an image, with comments explaining the optimization process. The code is as follows:

```
1 // =====  
2  
3 // =====  
4 // student_func.cu  
5 // =====  
6 // Angela Gross  
7 // CSCI-595: Parallel Programming  
8 // Homework 2  
9 // Spring 2015  
10 // =====  
11 // This blurs an image using CUDA and a square array of weights (see gaussian_blur() below for more details).  
12 //  
13 // In order to optimize:  
14 // - The channels weren't separated, (as suggested), and so only one call to a kernel was made  
15 // - Using this URL as a guide (http://cuda-programming.blogspot.com/2013/01/what-is-constant-memory-in-cuda.  
16 //   the filter was put into constant memory since it isn't ever changed  
17 // - Making sure that as much of the global memory is in local memory in the kernel  
18 // - Removing synchronization and error checks since it's done in the autograder (main.cpp) after it calls th  
19 //   method.  
20 //  
21 // Since the methods were prototyped in other source files used by the autograder, many functions have unnece  
22 // arguments or, like in the case of cleanup(), the function itself is unnecessary. (I wasn't able to actuall  
23 // these files to match on the Udacity assignment submission)  
24 // =====  
25
```

Below the code editor are four buttons: 'Reset', 'Test Run', 'Review Instructions', and 'Submit'. To the right of the 'Submit' button is an orange 'Continue' button. To the right of the code editor, there is a feedback section with a green thumbs-up icon and the text 'Correct!'. Below this, it says 'Please see detailed feedback in the output area below.' There is also a 'Get Help' section with several links and timestamps: 'Incorrect orientation of output. 1d ago', 'Need help cant figure whats wrong!!! 4d ago', 'CUDA error at: student_func.cu:329 6d ago', 'libdc1394 error - Help needed 7d ago', and 'Incorrect blur at the edges 7d ago'. At the bottom of the feedback section are two buttons: 'Start a Discussion' and 'Report an Issue'. At the bottom of the main content area, there is a message box that says: 'Good job! Your image matched perfectly to the reference image' and 'Your program ran and executed in 0.561248 ms.'